

L9b: American Options, CRR Pricing, and the Value of Early Exercise

CHEME 5660 Cornell University

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Objectives for Today

L9a priced contracts exercisable only at expiration. Today the holder gets a **choice of when to exercise**. We keep the same terminal payoffs, price both exercise styles on one lattice, and measure what that extra right is worth.

Today's concepts

- **Contrast European and American exercise**: explain why identical terminal payoffs can carry different premiums when one holder may exercise early.
- **Price both styles on one CRR lattice**: apply the European and American backward-induction rules under identical inputs and verify that the American value is never smaller.
- **Explain the premium difference**: separate intrinsic value, extrinsic value, and the early-exercise premium, and identify when the American premium is strictly higher.

Lectures and Examples

Lecture: the American exercise set, the CRR lattice, the two backward-induction rules, the three value differences, and the conditions under which American exercise is actually worth something.

Example 1: the two-step textbook put. Price both styles on one lattice and inspect the node where exercise beats continuation.

Example 2: an AMD option chain. Price the whole chain with CRR under both rules and compare with quoted bid-ask intervals.

Example 3: an NVDA call. Split the premium into intrinsic and extrinsic value across strike and time to expiration.

Same Terminal Payoff, Different Exercise Set

For strike $K > 0$ and underlying price $S \geq 0$, with $x^+ := \max(x, 0)$, the immediate-exercise values are:

$$h_c(S) = (S - K)^+, \quad h_p(S) = (K - S)^+.$$

- **European**: receives this payoff only at expiration T .
- **American**: may choose any exercise time $\tau \leq T$, which includes the strategy of never exercising early.

The American strategy set **contains** the European one, so under identical model inputs:

$$V_0^{\text{AM}} \geq V_0^{\text{EU}}.$$

The inequality need not be strict.

Build the Cox–Ross–Rubinstein Lattice

Split maturity T into N steps of length $\Delta t = T/N$. With continuously compounded risk-free rate g_y and annualized volatility $\sigma > 0$, CRR sets the move sizes and the pricing weight q :

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = e^{-\sigma\sqrt{\Delta t}}, \quad \boxed{q = \frac{e^{g_y\Delta t} - d}{u - d}}.$$

- This is **L3b's one-step result** for a continuous rate: there $R_f = e^{g_y\Delta t}$ and $q = (R_f - d)/(u - d)$.
- q is a **pricing weight**, not a forecast. L3b keeps p for the real-world up probability; only q appears today.
- No arbitrage requires $d < e^{g_y\Delta t} < u$, equivalently $0 < q < 1$. If that fails, **refine the step**, never clip q .
- With a continuous dividend yield δ , replace $e^{g_y\Delta t}$ in the numerator by $e^{(g_y - \delta)\Delta t}$.

One Lattice, Two Backward-Induction Rules

Index a node by its level $t \in \{0, \dots, N\}$ and its number of up moves $k \in \{0, \dots, t\}$, so $S_{t,k} = S_0 u^k d^{t-k}$, as in L3b and L4a. At expiration both styles agree, $V_{N,k} = h(S_{N,k})$, where h is whichever of h_c , h_p applies. For $t < N$ the discounted continuation value is:

$$C_{t,k} = e^{-g_y \Delta t} \underbrace{[qV_{t+1,k+1} + (1-q)V_{t+1,k}]}_{\text{pricing expectation}}.$$

The exercise convention changes **exactly one line**:

$$\boxed{V_{t,k}^{\text{EU}} = C_{t,k}^{\text{EU}}} \quad \boxed{V_{t,k}^{\text{AM}} = \max\{h(S_{t,k}), C_{t,k}^{\text{AM}}\}}.$$

Exercise where $h > C$, wait where $C > h$. Equality is indifference: the premium does not depend on the tie-break, the reported decision does.

A Two-Step Put Makes the Difference Visible

Take $S_0 = 50$, $K = 52$, two one-year steps, $u = 1.2$, $d = 0.8$, and $g_y = 5\%$, so $q \approx 0.6282$. Since $ud = 0.96 \neq 1$ this is a **generic** binomial tree, not a CRR calibration; the two recursions are unchanged either way.

At the one-year down node $S = 40$, exercising pays 12 while waiting is worth:

$$e^{-0.05}[0.6282(4) + 0.3718(20)] \approx 9.46.$$

The American holder **exercises**; the European holder must wait. Rolling both trees to the root:

$$V_0^{\text{EU}} \approx 4.19, \quad V_0^{\text{AM}} \approx 5.09.$$

The early-exercise right is worth about 0.90 USD/share here.

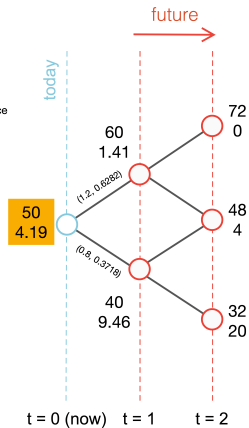
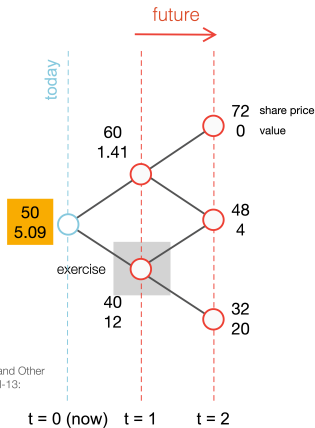
Example 1: `CHEME-5660-L9b-Example-Hull-American-Put-Fall-2026.ipynb`

One Node Carries the Whole Difference

Example: Hull 13.4 and 13.5

American Put contract

European Put contract



Reference: John C. Hull "Options, Futures, and Other Derivatives" 9th edition, 2015. Pearson, ISBN-13: 978-0133456318

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Only the down node differs: American 12, European 9.46.

Three Value Differences Answer Three Questions

Let V be the optimally managed value, h intrinsic value, and C continuation value. At an interior node, that is for $t < N$:

$$\underbrace{X_{t,k} = V_{t,k} - h(S_{t,k})}_{\text{extrinsic value}} = (C_{t,k} - h(S_{t,k}))^+ \geq 0,$$

because $\max\{h, C\} - h = \max\{0, C - h\}$. Continuation is undefined at expiration, where $V_{N,k} = h(S_{N,k})$ gives $X_{N,k} = 0$ directly. Across styles:

$$\underbrace{A = V^{\text{AM}} - V^{\text{EU}}}_{\text{early-exercise premium}} \geq 0.$$

- C : what is waiting worth?
- X : how far does the managed contract exceed immediate exercise?
- A : what does American flexibility add over the same European contract?

When Is the American Premium Strictly Higher?

- **Non-dividend calls, $g_y \geq 0$:** early exercise is never strictly better, so $V_c^{\text{AM}} = V_c^{\text{EU}}$. Paying K early and discarding optionality cannot help. For $g_y > 0$ waiting is strictly better; at $g_y = 0$ the holder can be indifferent.
- **Puts:** early exercise can be optimal. Receiving K sooner has financing value, but that has to **exceed the optionality given up**, which is why the put must also be sufficiently in the money.
- **Dividend-paying calls:** exercising before an ex-dividend date can pay when the dividend beats retaining the option and deferring K .
- **Negative rates, discrete dividends, funding spreads, constraints:** each needs its own model before any exercise rule follows.

American style guarantees only the **weak** inequality.

Examples 2 and 3: AMD chain premiums; NVDA intrinsic and extrinsic value

Optional Advanced Material

Advanced: [advanced/crr_factors/](#)

CHEME-5660-L9b-Advanced-CRR-Factors-Fall-2026.ipynb

Optional, not a prerequisite for L10a. Today's lecture **quotes** $u = e^{\sigma\sqrt{\Delta t}}$ and $d = e^{-\sigma\sqrt{\Delta t}}$; the notebook derives them. It writes the one-step log return as a two-point variable and matches its variance to the diffusion variance $\sigma^2\Delta t$, and it is explicit about the price: $ud = 1$ is a **normalization the modeler chooses**, not something the diffusion forces, the equal weights used to size the jump are **not** the pricing probability q , and the variance match holds only to leading order, discarding a term of order Δt^2 . It closes with the domain $|g_y|\sqrt{\Delta t} < \sigma$ that keeps $0 < q < 1$.

Notebook: [lectures/week-9/L9b/advanced](#), also linked from the lecture notebook.

Summary

Today we took L9a's European price as a benchmark, added the American exercise decision on the same lattice, and measured what that extra right is worth.

Key takeaways

- **American style expands the exercise set:** the American holder can always choose to wait until expiration, so the American premium is never smaller than the European premium under identical inputs.
- **Backward induction creates the difference:** the European recursion always continues, the American recursion takes the larger of intrinsic and continuation value, and that one line is the whole distinction.
- **Higher is not automatic:** on the AMD chain the American and European call premiums agree exactly, while the puts carry a positive early-exercise premium that reaches about 2.30 USD/share at the deepest in-the-money strike.

Next, L10a asks how a premium responds locally to spot, volatility, time, and interest rates through the Greeks.